Machine Learning Reveals How Molecular Structure and Environmental Conditions Jointly Drive PFAS Biotransformation

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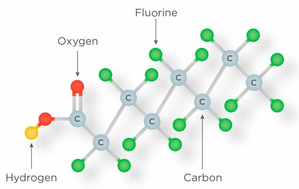
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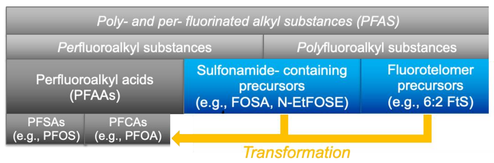
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Background

Per- and Polyfluoroalkyl Substances (PFAS)

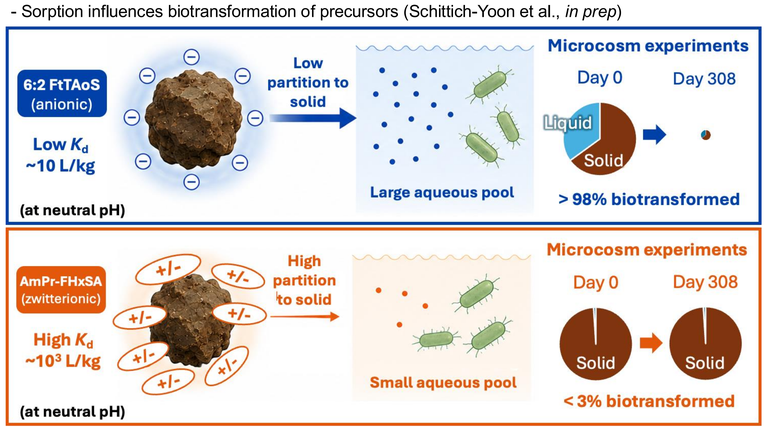
- PFAS are often called "forever chemicals" due to their extreme persistence in the environment
- Uses of PFAS: firefighting activities, water- and oil-resistant fabrics, and semiconductor manufacturing
- Toxicity:
  - Acute toxicity to aquatic species (Jones et al., 2022)
  - Increased cancer risk (Fenton et al., 2021)
  - Reduced immune system function (Kielsen et al., 2015)





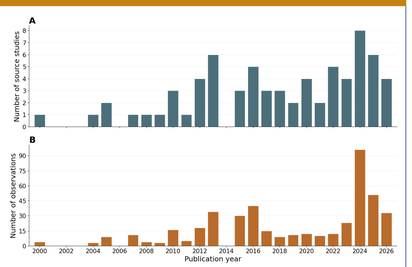
- More than half of currently used PFAS are precursors, largely due to the phaseout of terminal PFAAs (Adamson et al. 2020)
- Precursors themselves can also exhibit toxicity (Zhang et al., 2021)
- Diverse charge distributions → Different sorption behaviors (Yoon et al., 2025)

- Sorption influences biotransformation of precursors (Schittich-Yoon et al., *in prep*)



Objectives

- Understanding the role of environmental factors (e.g., solid-to-liquid ratio) in PFAS biotransformation potential across 70 studies
  - ***Hypothesis 1*:** Incorporating PFAS-specific and environmental descriptors improves prediction of PFAS biotransformation outcomes across heterogeneous experimental systems vs. general molecular descriptors alone
  - ***Hypothesis 2*:** The effects of environmental conditions on biotransformation differ among PFAS subclasses and depend on structural characteristics such as molecular charge and lipophilicity



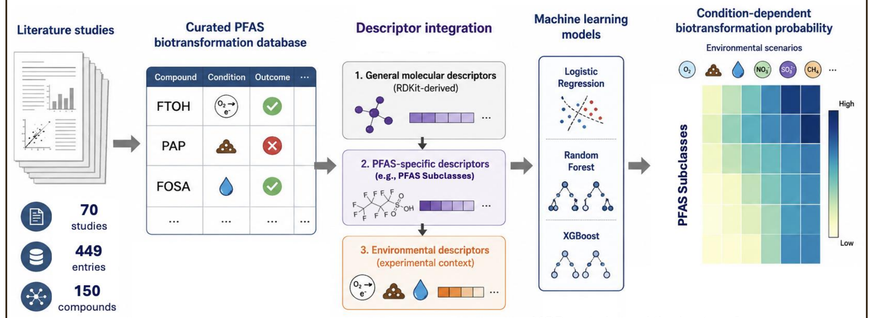
Publications

1. **Hyun Yoon**, Fuhar Dixit, Jonathan J. Uhler, Anna-Ricarda Schittich, Edmund H. Antell, Lisa Alvarez-Cohen and David L. Sedlak. 2025. Effects of organic matter, iron (hydr)oxides, and iron reductive dissolution on PFAS sorpti on in AFFF-impacted solids. ***Environmental Science & Technology Letters***, 13(1), 130-136.

2. Anna-Ricarda Schittich†, **Hyun Yoon†**, Leyan Peng, Fuhar Dixit, Tae-Kyoung Kim, Fanshu Geng, Damian Helblin g, Christopher Olivares, David L. Sedlak, and Lisa Alvarez-Cohen. Charge-Dependent Sorption Controls the Bio availability and Biotransformation of PFAS Precursors in AFFF-Impacted Solids. *In prep.* (†Equal first author cont ribution)

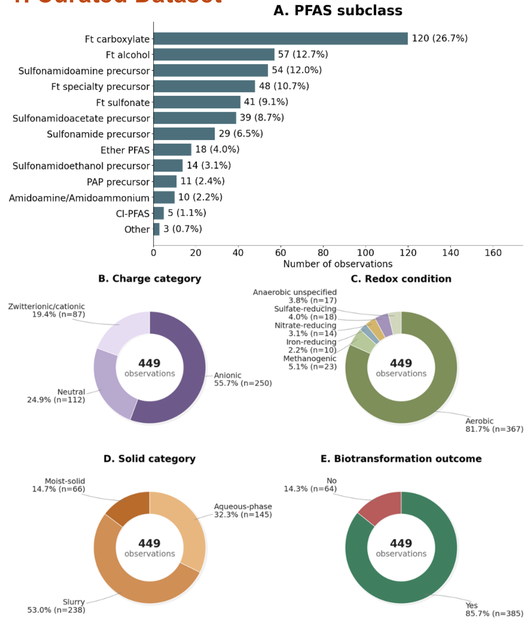
Methodology & Results

Research Pipeline

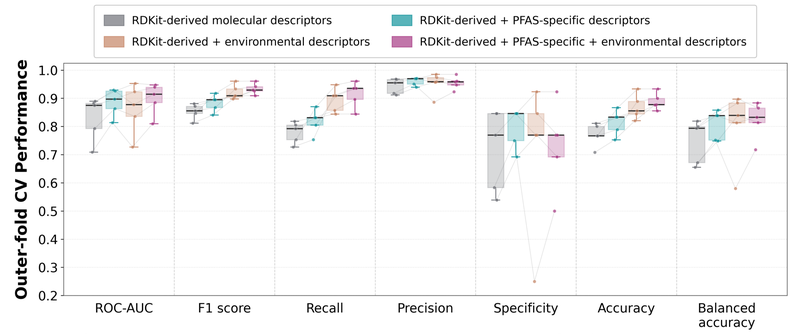


XGBoost showed the best performance; and was selected for downstream analysis!

1. Curated Dataset

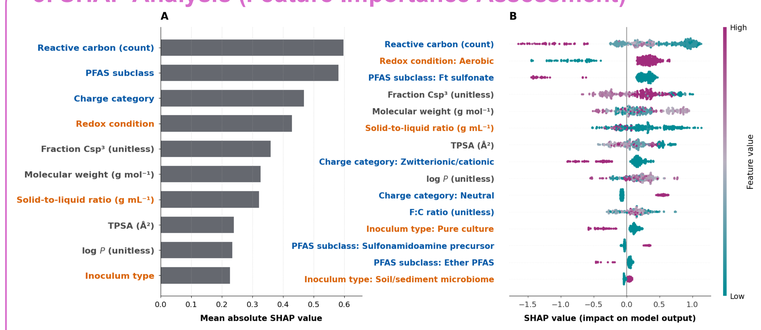


2. Model Performance Metrics (XGBoost)

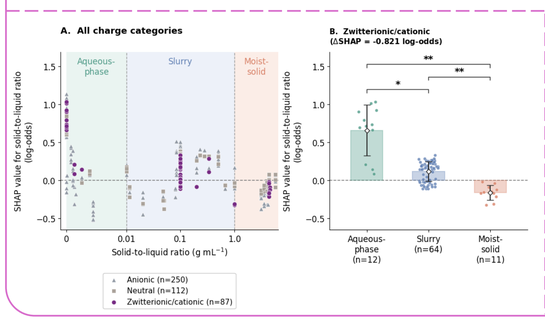


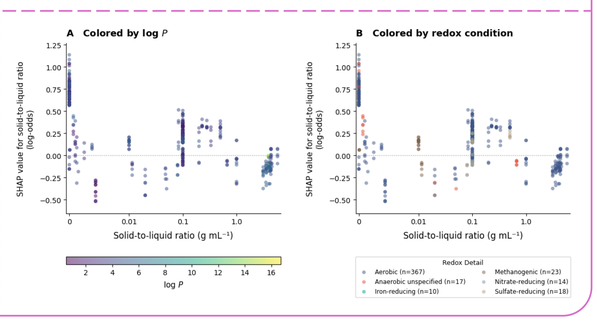
Adding PFAS-specific and environmental descriptors improved most performance metrics. Specificity remained more variable, consistent with the limited number of negative observations (14.3%)

3. SHAP Analysis (Feature Importance Assessment)

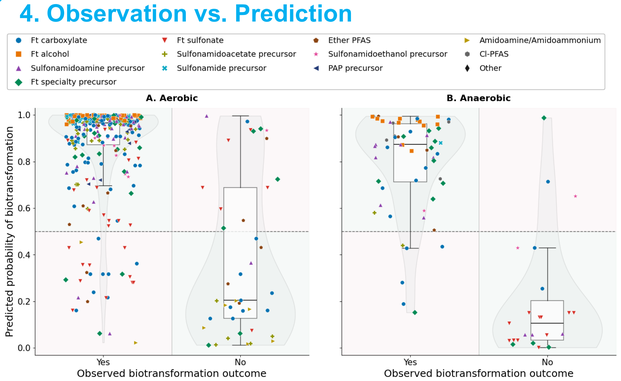


- SHAP analysis identified the most influential descriptors for predicting PFAS biotransformation
- Beeswarm plot (upper-left panel B) highlights key feature impacts (e.g., positive: aerobic conditions, neutral structures; negative: zwitterionic/cationic charges, pure culture systems)
- SHAP interaction dependence plots (lower-left panels) reveal that higher solid-to-liquid ratios were associated with more negative SHAP contributions, particularly for zwitterionic/cationic PFAS
- High lipophilicity (log *P*) at high S:L ratio also resulted in lower SHAP values
- Aerobic condition was evaluated across a wide range of S:L ratios, whereas anaerobic conditions were mostly studied at low S:L ratios





4. Observation vs. Prediction



- XGBoost Model Robustness: High Recall was maintained across both aerobic (0.917) and anaerobic (0.898) regimes at the 0.5 threshold.
- Redox Shifts: Positive observations under aerobic conditions were concentrated at high predicted probabilities (>0.85), whereas anaerobic predictions were lower and more variable. Non-transformed anaerobic observations clustered at low probabilities (<0.20).
- Charge Effects: Neutral precursors showed near-universal biotransformation (98.2%), while zwitterionic/cationic species had the highest non-transformation rate (32.2% vs. 13.6% for anionic).

Environmental Implications & Conclusion

Environmental Implications

- The model supports condition-specific screening of PFAS biotransformation within the literature-represented domain (Figure on the right)
  - E.g., Ft alcohols retained high predicted transformation probabilities under both aerobic and anaerobic scenarios, whereas other precursors show lower transformation probabilities under anaerobic conditions

Conclusion

- Integrating PFAS-specific and environmental descriptors significantly improves predictive performance for biotransformation outcomes
- The influence of environmental descriptors (e.g., S:L ratios, redox conditions) on PFAS biotransformation varies by PFAS structures, such as molecular charge and lipophilicity (log *P*)
- The curated dataset and ML-derived predictions inform and bridge critical knowledge gaps (compounds × conditions)

